FIITJEE Students From Classroom / Integrated School Programs have secured to 2 Zonal, 6 State & 18 City Topper Ranks. 33 in Top 100, 78 in Top 200 and 205 in Top 500 All India Ranks bagged by FIITJEE Students from All Programs have qualified in JEE (Advanced), 2015.

ALL INDIA TEST SERIES

FIITJEE **JEE(Advanced)-2016**

ANSWERS, HINTS & SOLUTIONS

FULL TEST – I
(PAPER-1)

Q. No.	PHYSICS	CHEMISTRY	MATHEMATICS
1.	C	D	B
2.	C	D	C
3.	B	A	D
4.	B	C	A
5.	B	C	D
6.	C	C	A
7.	B	A	C
8.	C	A, B, C	A, B, D
9.	A, B, C, D	B	A, B, C
10.	A, B, C	B, C	B, C
11.	A, B, C	A, C	A, B
12.	A	C	C
13.	B	B	D
14.	C	A	D
15.	D	A	B
16.	A	D	A
1.	(A) → s, (B) → pt, (C) → q, (D) → pr	(A) → r, (B) → pt, (C) → qs, (D) → qs	(A) → prt, (B) → q, (C) → t, (D) → s
2.	(A) → pqt, (B) → prt, (C) → s, (D) → qr	(A) → pqs, (B) → q, (C) → rt, (D) → q	(A) → qt, (B) → q, (C) → prs, (D) → t
1.	3	4	1
2.	5	7	2
3.	9	3	1
4.	9	3	5
5.	2	2	7
6.	4	3	2

Physics

PART – I

SECTION – A

1. C

$\tan^{-1}\left(\frac{1}{e}\tan\theta\right)$

$\because$ vertical component after collision $= eu\cos\theta$

& Horizontal component $= u\sin\theta$

$\Rightarrow \tan\phi = \frac{u\sin\theta}{eu\cos\theta} \Rightarrow \phi = \tan^{-1}\left(\frac{1}{e}\tan\theta\right)$

2. C

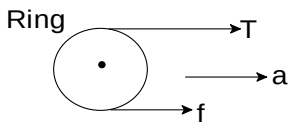
Zero

$\because T + f = 2a$

$TR - fR = 2R^2\alpha$

$a = \alpha R$

$\Rightarrow f = 0.$



3. B

$\because I_1 = \frac{V}{R}e^{-t/RC_1}$ & $I_2 = \frac{V}{R}e^{-t/RC_2}$

$\Rightarrow \frac{I_1}{I_2} = e^{\frac{t}{R}\left(-\frac{1}{C_1}+\frac{1}{C_2}\right)}$

$\Rightarrow \frac{I_1}{I_2} = e^{\frac{t}{RC_1}}$

4. B

$\because f = \frac{1}{2\pi\sqrt{LC}}, \quad f + 50 = \frac{1}{2\pi\sqrt{LC/K}}$

$\Rightarrow K = \left(\frac{f+50}{f}\right)^2 \Rightarrow K = \left(1 + 2\times\frac{50}{f}\right) = \left(1+\frac{100}{10^4}\right)$

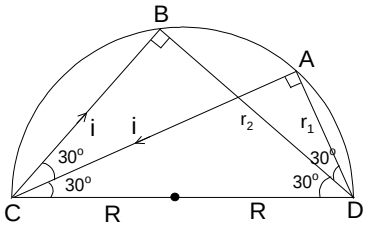
$\Rightarrow K = 1.01$

5. B

$\because r_1 = 2R\sin 30^o, r_2 = 2R\sin 60^o = R\sqrt{3} = R$

$B_D = B_{AC} + B_{CB} = \left(\frac{\mu_0 i}{4\pi r_1}\sin 60^o - \frac{\mu_0 i}{4\pi r_2}\sin 30^o\right)$

$\Rightarrow B_D = \frac{\mu_0 i}{4\pi R\sqrt{3}}$



6. C

$\text{Beats} = \frac{V}{4\ell} - \frac{V}{4(\ell+\Delta\ell)} = \frac{V\Delta\ell}{4\ell(\ell+\Delta\ell)}$

$= \frac{V\Delta\ell}{4\ell^2}$

7. B

$\frac{3}{2}\sin 60^\circ \geq \mu \sin 90^\circ$

$\mu \leq \frac{3}{2}, \frac{\sqrt{3}}{2}$

$\mu \leq \frac{3\sqrt{3}}{4}$

8. C

$\because -13.6 + 10.2 = -3.4\text{eV} \longrightarrow$ 1st collision

& $-3.4 + 15 = 11.6\text{eV} \longrightarrow$ 2nd collision.

9. A, B, C, D

$V \propto \frac{1}{n}$

$\text{P.E.} \propto (-)\frac{1}{r}$

$T \propto r^{3/2}$

$\text{T.E.} \propto (-)\frac{1}{r}$

10. A, B, C

$\because V_s^2 = (3)^2 + (8-4)^2 = 25 \Rightarrow V_s = 5$ volt

$Z = \frac{Vs}{I} = 5\Omega$

$V_R = 3 = 1 \times R \Rightarrow R = 3\Omega$

$\text{P.F.} = \frac{R}{Z} = \frac{3}{5} = 0.6$

$\because V_L > V_C \Rightarrow I$ lags V.

11. A, B, C

$\because \frac{1}{f} = \frac{1}{v} - \frac{1}{u} \Rightarrow \frac{v}{f} = \left(1 - \frac{v}{u}\right)$

since m is positive in the graph,

$\therefore m = \left|\frac{v}{u}\right| = (-)\frac{v}{u}$

$\Rightarrow m = \left(\frac{v}{f} - 1\right) = (-)\frac{v}{u}$

$\therefore$ intercept on $m-$axis $= -1$ & $\tan\theta = \frac{1}{f}$

12. A

$\because \omega = \frac{V}{R}, \text{OP} = \sqrt{x^2 + y^2}$

origin is instantaneous centre of rotation

$\therefore V_P = \omega r = \omega(\text{OP}) = \frac{V}{R}\sqrt{x^2 + y^2}$

13. B

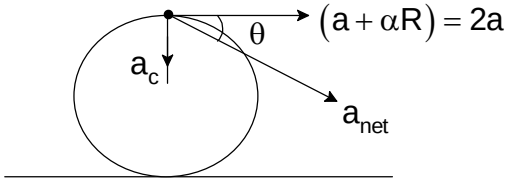
$\because a = \alpha R$

$a_c = \omega^2 R = \left(\frac{V}{R}\right)^2 R = \frac{V^2}{R}$

$= (2a)$

$\tan\theta = \frac{a_c}{2a} = 1$

$\therefore \theta = 45^o$



14. C

Thermal heat capacity, $C = \frac{Q}{\Delta T} = \frac{300}{20} = 15 J/^oC$

15. D

Specific heat capacity, $S = \frac{Q}{m\Delta T} = \frac{300}{0.025 \times 20} = 600 J/kg\ ^oC$

16. A

Molar heat capacity

$= \frac{Q}{n\Delta T} = \frac{300}{\left(\frac{25}{50}\right)(20)}$

$= 30 J/mol\ ^oC$

SECTION – B

1. $(A \to s)$ $(B \to p, t)$ $(C \to q)$ $(D \to p, r)$

In wrist watch spring force is not dependent on gravity.

$T = 2\pi\sqrt{\frac{m}{K}}$

2. $(A \to p, q, t)$ $(B \to p, r, t)$ $(C \to s)$ $(D \to q, r)$

In nuclear reactions energy, charge, mass and momentum remain conserved.

SECTION – C

1. 3

for R^{max}, $\theta = 45^o \Rightarrow H^{max} = \frac{(20)^2 \sin^2(45^o)}{2 \times 10} = 10m$

$\because H^{max}$ allowed $= 5m$,

$\Rightarrow 5 = \frac{(20)^2 \sin^2\theta}{2 \times 10} \Rightarrow \theta = 30^o$

$\Rightarrow R^{max} = \frac{(20)^2 \sin 60^o}{10} m = 20\sqrt{3}m.$

2. 5

$\because M_{cylinder} = \left(\pi R^2 \times 3R\right)\rho.\ I_{cylinder} = \frac{1}{2}\left(3\pi R^3\rho\right)R^2$

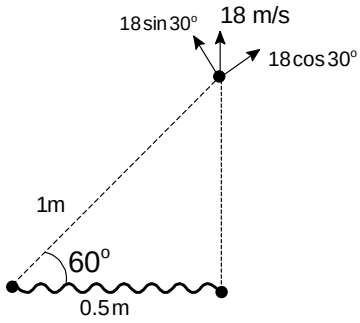
$M_{sphere} = \frac{2}{3}\pi R^3\rho,\ I_{sphere} = \frac{2}{5}\left(\frac{2}{3}\pi R^3\rho\right)R^2$

$\therefore I_{remaining\ portion} = I_{cylinder} - 2\ I_{sphere} = \frac{29}{30}\pi R^5\rho.$

3. 9

$\because 18\sin 30^o = \omega r = \omega \times 1$

$= \omega = 9\ m/s.$



4. 9

$\because \frac{1}{F} = \frac{1}{f_1} + \frac{1}{f_2}$

$\Rightarrow \frac{1}{24} = \left[\left(\frac{\mu_1}{\frac{4}{3}} - 1\right)\left(\frac{1}{20} + \frac{1}{20}\right) + \left(\frac{\mu_2}{\frac{4}{3}} - 1\right)\left(\frac{1}{-20} - \frac{1}{20}\right)\right]$

$\Rightarrow (\mu_1 - \mu_2) = \left(\frac{5}{9}\right).$

5. 2

$\because I = \frac{V}{R} = \frac{200}{80} = 2.5A \ \& \ P = \frac{V^2}{R} = \frac{(200)^2}{80} = 500\,watt$

When the wire is divided into two equal parts and connected in parallel we get maximum power.

$\therefore$ Resistance of each part $= 40\Omega$

$\therefore I^{max} = 5A$

$\Rightarrow P_{max} = \frac{V^2}{R_{eq}} = \frac{(200)^2}{(20)} = 2000W = 2kw.$

6. 4

$\because \Delta V_{AP} = \frac{Q}{3}, \Delta V_{PB} = \frac{Q}{6}$

$\because 12 = \frac{Q}{3} + \frac{Q}{6} \Rightarrow Q = 24\mu C$

$\Rightarrow \Delta V_{AP} = \frac{24}{3} = 8$

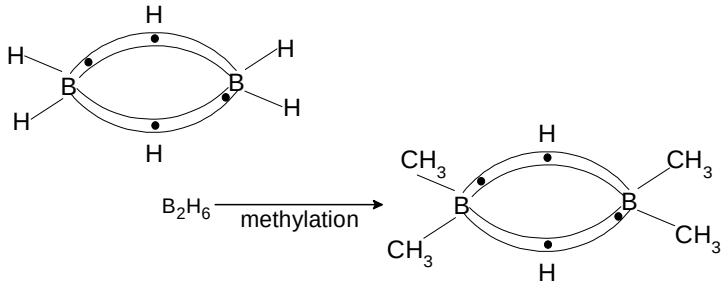
$\Rightarrow V_A - V_P = 8 \Rightarrow V_P = 4volt.$

Chemistry

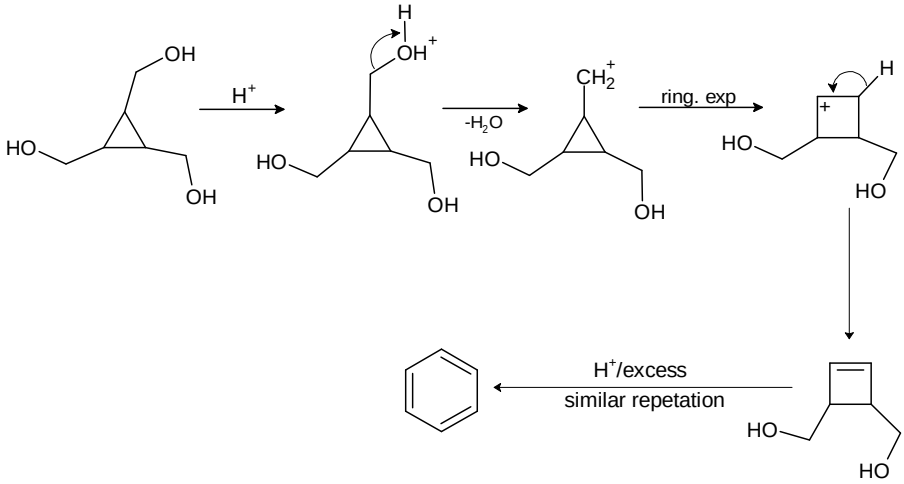
PART – II

SECTION – A

1. D



2. D



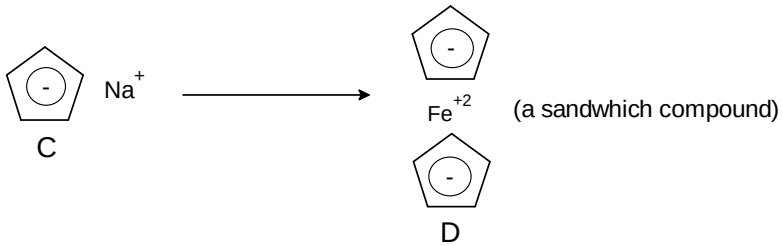
3. A

At nodal surface, $\psi = 0$

$\Rightarrow r = \frac{2}{z} = \frac{2}{z}$ Å = 1Å

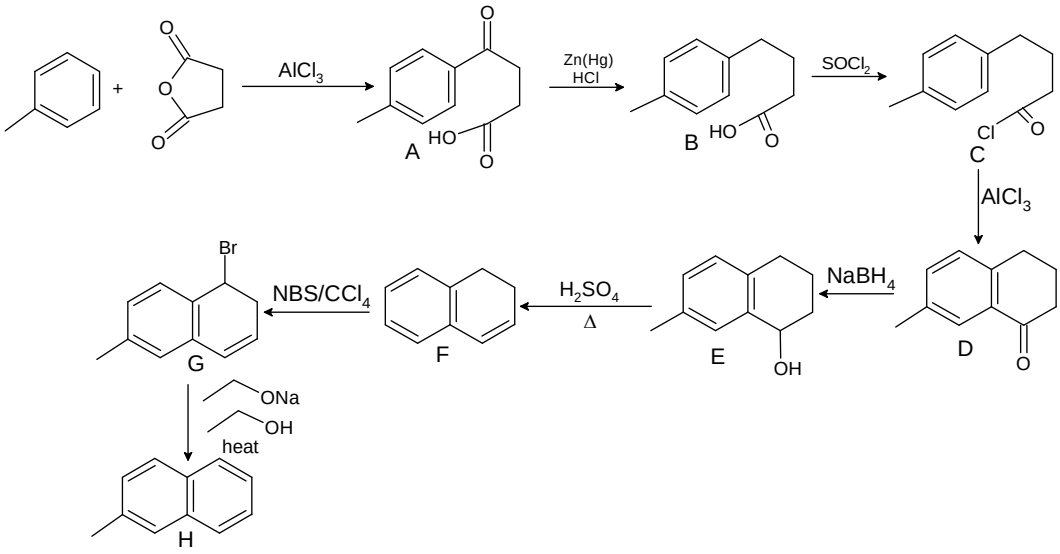
4. C

Co – ordination no of Fe^{+2} in D is 6

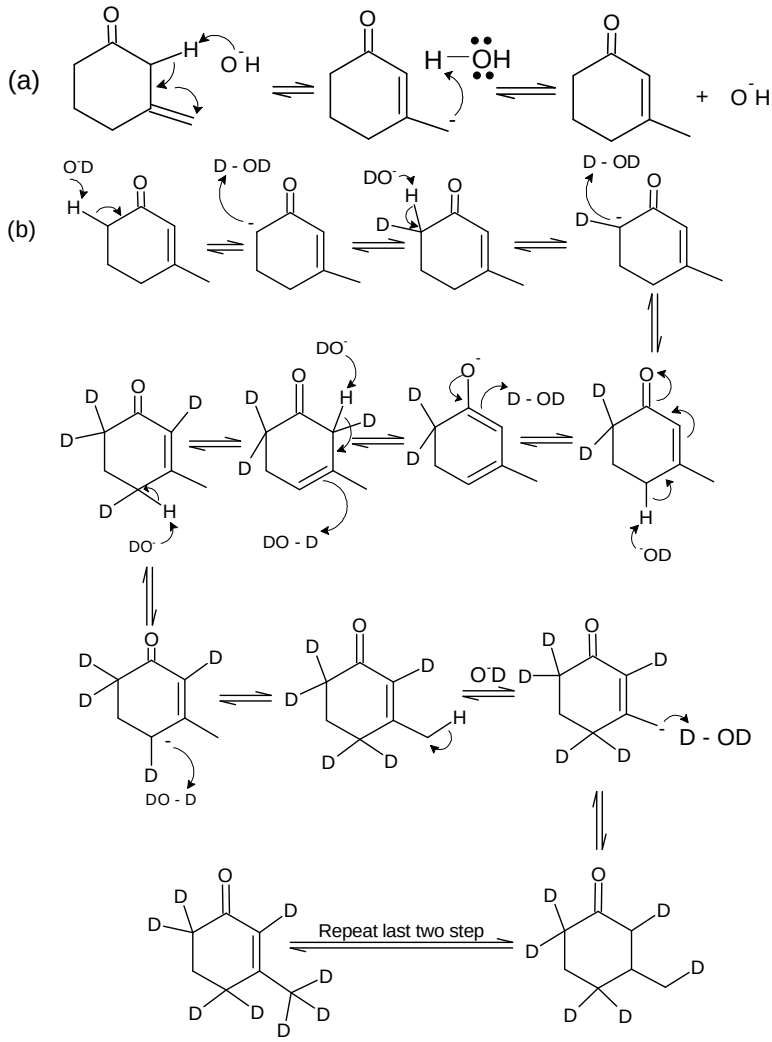


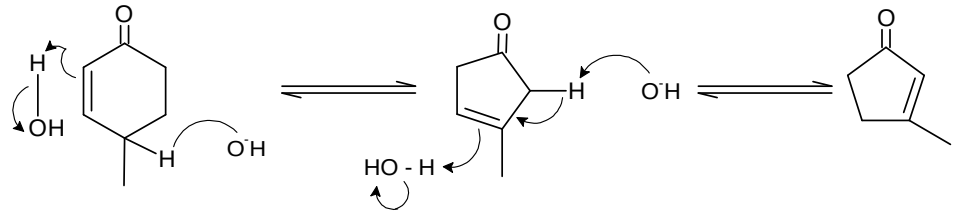
(a sandwhich compound)

5. C


AlCl3
Zn(Hg)
HCl
SOCl2
AlCl3
NaBH4
H2SO4
Δ
NBS/CCl4
ONa
OH
heat
A
B
C
D
E
F
G
H

6. C


(a)
(b)
D - OD
DO - D
Repeat last two step

(d) 

7. A

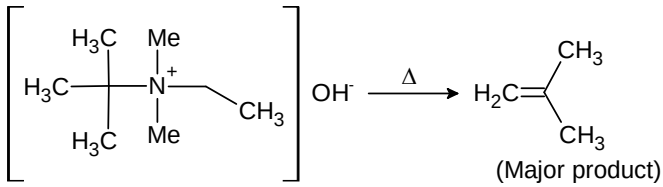
Synergic bonding.

8. A, B, C

$$\log\frac{K_2}{K_1}=\frac{\Delta H^o}{2.303R}\left[\frac{T_2-T_1}{T_1T_2}\right]\ K_1\&K_2 \text{ are eq. constant}$$

$$\log\frac{K_2}{K_1}=\frac{E_a}{2.303R}\left[\frac{T_2-T_1}{T_1T_2}\right]\ K_1\&K_2 \text{ are rate constant}$$

9. B



(Major product)

Due to greater probability of losing β-H (9 : 3 ratio).

10. B, C

(b) conjugate base of $H_2PO_4^-$ is HPO_4^{2-}

(c) pH of 0.1 M NaCl (aqueous solution) = $\frac{1}{2}pK_w$

11. A, C

$A^{+3}+3e^-\longrightarrow Al$

Number of moles of Al produced $=\frac{4.5}{27}=0.166$

Number of moles of electrons used $=3\times0.166$

$H^++e^-\longrightarrow\frac{1}{2}H_2$

Number of moles of H_2 produced $=\frac{1}{2}\times3\times0.166$

Volume of H_2 produced $=1.5\times0.166\times22.4\ell=5.57\ell=5.6\ell$

12. C

2nd step is r.d.s. in sulphonation reaction.

13. B

2nd step is fast step in nitration.

14. A

$\left(P+\frac{a}{V}\right)(V-b)=RT$

on converting into virial equation from, we get

$$PV=\left[RT\frac{\left(b-\frac{a}{RT^2}\right)}{V}+\frac{b^2}{V^2}+\frac{b^3}{V^4}+.....\right] \text{ to higher power of n} \quad \ldots(i)$$

For Boyle's temperature, 2nd virial co-efficient becomes zero

$\therefore b=\frac{a}{RT^2}$

$T_B=\sqrt{\frac{a}{Rb}}$

$\therefore T_B^2=\frac{a}{RB}$

$\therefore T_i=\frac{2a}{Rb}\Rightarrow T_i=2T_B^2$

15. A

Let Boyle's temperature, $T_B^{'}$

$\therefore T_B=\sqrt{\frac{a}{Rb}}$

$T_B^{'}=\sqrt{\frac{8a}{R\times 2b}}=2\sqrt{\frac{a}{Rb}}=2T_B$

16. D

$b^2 = 0.04$

b = 0.2 lit/mol

$V_{real} = 4 \times 22.4 + 4 \times 0.2 = 90.4$ lit

SECTION – B

1. $(A\to r),(B\to p,t),(C\to q,s)(D\to q,s).$

(A) Sic covalent carbide (Not hydrolysed)

(B) $Al_4C_3+12H_2O\longrightarrow 4Al(OH)_3+3CH_4$

(C) $CaI_2+2H_2O\longrightarrow Ca(OH)_2+H-C\equiv C-H$

(D) $Mg_2C_3+4H_2O\longrightarrow 2Mg(OH)_2+CH_3-C\equiv CH$

2. $(A \to p,q,s),(B \to q),(C \to r,t)(D \to q).$

$Al_2(SO_4)_3 \longrightarrow 2Al^{+3} + 3SO_4^{--}$

0.2M 0.3M

$AlPO_4 \longrightarrow Al^{+3} + PO_4^{-3}$

0.1M 0.1M

Urea $\longrightarrow$ Urea

0.1M

$MgCl_2 \longrightarrow Mg^{+2} + 2Cl^{-}$

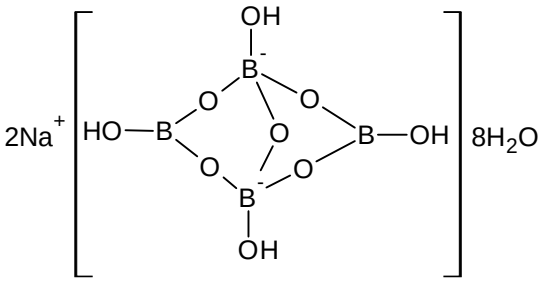
0.1M 0.2M

SECTION – C

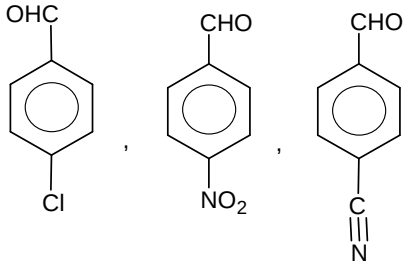
1. 4

Based on fact

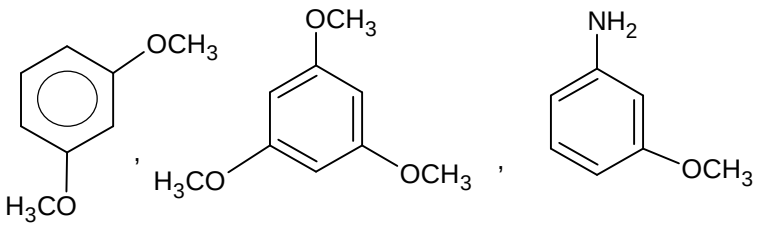
2. 7



3. 3



4. 3



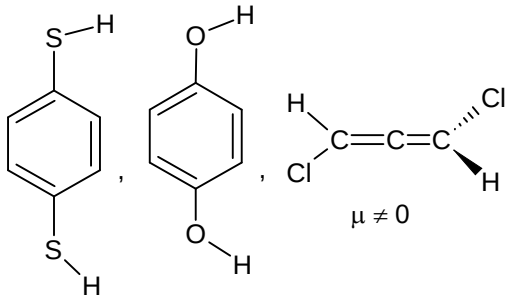
5. 2

$$t_{99} = \frac{1}{K}\ell n100; \quad t_{90} = \frac{1}{K}\ell n\frac{100}{10}$$

$$\frac{t_{99}}{t_{90}} = \frac{\ell n100}{\ell n10} = \frac{2}{1} = 2$$

hence $x = 2$

6. 3



$\mu \neq 0$ $\mu \neq 0$ $\mu \neq 0$

Mathematics

PART – III

SECTION – A

1. B

$\left|\overrightarrow{PQ}\right| = \sqrt{r^2+1-2r\cos\frac{\pi}{3}} = \sqrt{r^2+1-r}$

& $\left|\overrightarrow{AT}\right| = \sqrt{r^2+1-2r\cos\theta}$

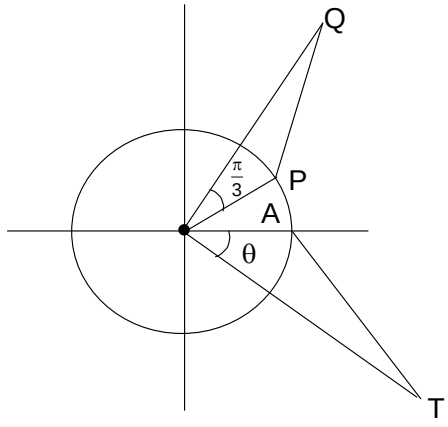
Where 'Q' is the angle between $\overrightarrow{OA}$ and $\overrightarrow{OT}$.

now, $\left|\overrightarrow{PQ}\right|^2 = 4\left|\overrightarrow{AT}\right|^2$

$\Rightarrow \left(1+r^2-r\right) = 4\left(r^2+1-2r\cos\theta\right)$

$\cos\theta = \frac{3r^2+r+3}{8r} \Rightarrow -1 \le \frac{3r^2+r+3}{8r} \le 1; \quad r>0$

$\Rightarrow r \in \left[\frac{7-\sqrt{13}}{6}, \frac{7+\sqrt{13}}{6}\right].$



2. C

Here two balls are drawn and the result is known that the two balls are white, therefore Bayes's theorem should be used.

Required probability = $\frac{\frac{1}{4}.1}{\frac{1}{4}.1+\frac{1}{4}.\frac{6}{10}+\frac{1}{4}.\frac{3}{10}+\frac{1}{4}.\frac{1}{10}} = \frac{1}{2}$

3. D

Let $Q(x) = f(x)-g(x)$ in $[x_0 x]$

By LMVT,

$Q(x)-Q(x_0) = Q'(x)-Q'(x_0)$ for some $C \in (x_0, x)$

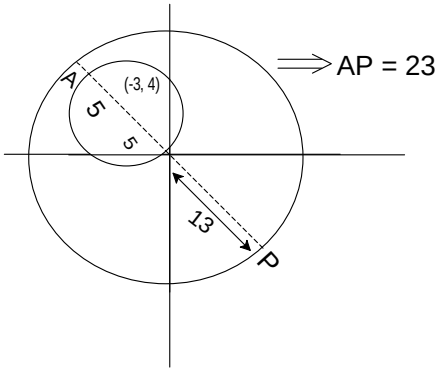
$\because Q(x_0) = 0,\ Q'(C) = f'(C)-g'(C) > 0$

$\Rightarrow Q(x)-Q(x_0) = Q(x)$

$= \left[F'(C)-g'(C)\right](x-x_0) > 0$

$\Rightarrow Q(x) > 0 \Rightarrow f(x) > g(x).$

4. A



5. D

$$S_n = \frac{5}{4} - \frac{2n+5}{2(n+1)(n+2)}$$

$$\Rightarrow K = \frac{5}{4}$$

6. A

Equation of tangent at 'P' is

$$Y - y = m(X - x)$$

Now, $\left|\frac{y - mx}{\sqrt{1+m^2}}\right| = x$

$$\Rightarrow \frac{y^2 - x^2}{2xy} = \frac{dy}{dx} \qquad \text{putting } y = vx$$

Solving Differential equation

$$x^2 + y^2 = Cx$$

It passes through $(1,2) \Rightarrow C = 5$

Equation of curve $\equiv x^2 + y^2 - 5x = 0$

7. C

$$T \equiv \frac{x\cos\theta}{3} + \frac{y\sin\theta}{2} = 1$$

$x = 0,\ y = 2\operatorname{cosec}\theta,$

chord A'P;

$$y = \frac{2\sin\theta}{3(\cos\theta+1)}(x+3)$$

$$x = 0,\ OM = \frac{2\sin\theta}{(\cos\theta+1)}$$

$$\text{Now, } OQ^2 - MQ^2 = OQ^2 - (OQ - OM)^2$$

$$= 2(OM)(OQ) - OM^2$$

$$= OM[2OQ - OM]$$

$$= \frac{2\sin\theta}{(\cos\theta+1)}\left[\frac{4}{\sin\theta} - \frac{2\sin\theta}{1+\cos\theta}\right]$$

$$= \frac{2\sin\theta\left[4 + 4\cos\theta - 2\sin^2\theta\right]}{(1+\cos\theta)^2\sin\theta}$$

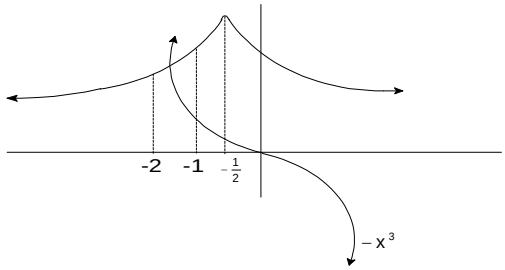
$$= \frac{4\sin\theta\left[2 + 2\cos\theta - 1 + \cos^2\theta\right]}{(1+\cos\theta)^2\sin\theta}$$

$$= 4$$

8. A, B, D

$$f(x) = 0 \Rightarrow x^3(x^2+x+1) + (x^2+x+1) + 2 = 0$$

$$\Rightarrow -x^3 = 1 + \frac{2}{x^2+x+1}$$



9. A, B, C

Take $f(x)=[x]$ and $g(x)=\{x\}$ and proceed.

10. B, C

$$I=\int \tan^{-1}\left(\frac{\cos 2x}{1+\sin 2x}\right)dx=\int \tan^{-1}\left[\tan\left(\frac{\pi}{4}-x\right)\right]dx$$

$$=\int\left(\frac{\pi}{4}-x\right)dx$$

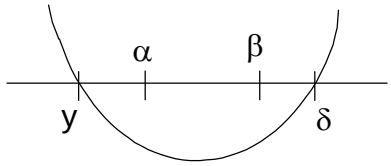
$$I=\frac{\pi}{4}x-\frac{x^2}{2}+C\Rightarrow g(x)=\frac{\pi}{4}x-\frac{x^2}{2}.$$

11. A, B

$$\alpha,\beta=\frac{2\pm\sqrt{4-4\left(a^2-1\right)}}{2}=1+a,1-a$$

$$\Rightarrow f(1-a)<0\ \&\ f(1+a)<0$$

$$\Rightarrow a\in\left(-\frac{1}{4},1\right)$$



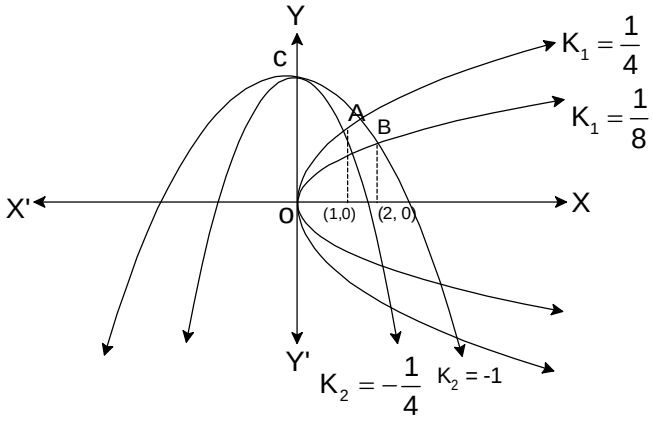
12. C

The value of K_1 & K_2 decide the opening of parabola

$\Rightarrow A_{max}=\text{Area}(OBCO)$

$\Rightarrow A_{min}=\text{Area}(OACO)$

13. D



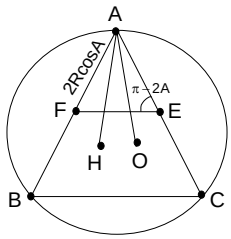
14. D

$$\text{Area}(BFEC)=\text{Area}(\Delta ABC-\Delta AFE)$$

$$=\text{or}\ (\Delta AOB+\Delta BOC+\Delta COA-\Delta AFE)$$

$$=\frac{1}{2}R^2\left[\sin 2C+\sin 2A+\sin 2B-\sin(\pi-2A)\right]$$

$$=\frac{1}{2}R^2(\sin 2B+\sin 2C)$$



15. B

$AE = R, AF = 2R\cos A$

$\Rightarrow EF = R$

16. A

$\angle AFE = \frac{\pi}{3}$

$\Rightarrow \Delta AFE$ is equilateral

$\Rightarrow$ sum of square of altitudes $= 3\left(\frac{R\sqrt{3}}{2}\right)^2 = \frac{9}{4}R^2.$

SECTION – B

1. (A $\longrightarrow$ p, r, t) (B $\longrightarrow$ q) (C $\longrightarrow$ t) (D $\longrightarrow$ s)

$(A)\tan\left(\frac{\theta_1+\theta_2+\theta_3+\theta_4}{2}\right) = 0 \Rightarrow \theta_1+\theta_2+\theta_3+\theta_4 = 2n\pi$

$(B)x_2 = 4 \Rightarrow t^2 = 2$, TR is diameter $= 6$

Area of circle $= \frac{\pi}{4}(TR)^2 = 9\pi$

$(C)6$

$(D)\left(y_A + y_B + y_C\right)^2 = \left(\sqrt{3}\right)^2 = 3$

2. (A $\longrightarrow$ q, t) (B $\longrightarrow$ q) (C $\longrightarrow$ p, r, s) (D $\longrightarrow$ t)

SECTION – C

1. 1

D.C's of OP are,

$= \left(\frac{1}{\sqrt{3}}, \frac{1}{\sqrt{3}}, \frac{1}{\sqrt{3}}\right)$

$\Rightarrow$ D.C's of $AL \equiv \left(-\frac{1}{\sqrt{3}}, \frac{1}{\sqrt{3}}, \frac{1}{\sqrt{3}}\right)$

D.C's of $BM \equiv \left(\frac{1}{\sqrt{3}}, -\frac{1}{\sqrt{3}}, \frac{1}{\sqrt{3}}\right)$

D.C's of $CN \equiv \left(\frac{1}{\sqrt{3}}, \frac{1}{\sqrt{3}}, -\frac{1}{\sqrt{3}}\right)$

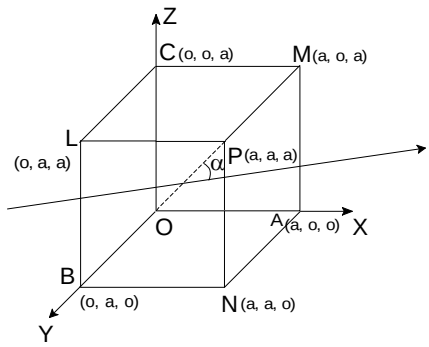
Let D.C's of required line be (ℓ, m, n)

$\therefore \cos\alpha = \frac{\ell+m+n}{\sqrt{3}}, \cos\beta = \frac{\ell+m-n}{\sqrt{3}}$

$\cos y = \frac{\ell+m+n}{\sqrt{3}}, \cos\delta = \frac{\ell+m-n}{\sqrt{3}}$

Hence, $\cos^2\alpha + \cos^2\beta + \cos^2 y + \cos^2\delta = \frac{4}{3}\left(\ell^2+m^2+n^2\right) = \frac{4}{3}$

$\Rightarrow \left[\cos^2\alpha + \cos^2\beta + \cos^2 y + \cos^2\delta\right] = \left[\frac{4}{3}\right] = 1.$



2. 2

The gives equation is $(x-1)(y-2)=4$

$$XY=4$$

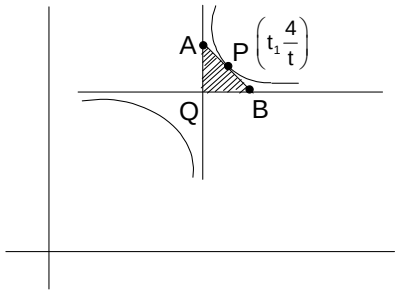
Equationof tangent is,

$$Y-\frac{4}{t}=-\frac{4}{t^2}(x-t)$$

$$\Rightarrow A\equiv\left(0,\frac{8}{t}\right)\ \&\ B\equiv(2t_1 0)$$

$$\text{Area}=\frac{1}{2}\times\frac{8}{t}\times 2t=8$$

$$4K=8\Rightarrow k=2.$$



3. 1

$$\frac{CE}{DE}=\frac{(a+b)^2}{C^2}=\frac{(2+3)^2}{16}=\frac{25}{16}$$

$$\Rightarrow\left[\frac{CE}{DE}\right]=1$$

4. 5

$$\sum_{K=1}^{10}\int_0^1 f(K-1+x)dx=\int_0^1 f(x)dx+\int_0^1 f(x+1)dx.....\int_0^1 f(x+9)$$

$$=\text{putting } x+1=t,(x+2)=t.....(x+9)=t$$

$$\Rightarrow\int_0^1 f(x)dx+\int_1^2 f(x)dx+\int_2^3 f(x)dx.....\int_9^{10} f(x)dx$$

$$=\int_0^{10} f(x)dx=5.$$

5. 7

Every ball have two options

$\Rightarrow$ 4 balls can be put in 2^4 ways.

Arrangement among themselves in 2!

$$=\frac{1}{2!}\times 2^4=2^3$$

But, above count also includes the one case in which all the balls are put in one box,

$\Rightarrow$ Required number of ways $= 2^3 - 1 = 7$.

6. 2

$$\begin{vmatrix}(K+1)^3 & (K+2)^3 & (K+3)^3\\ (K+1) & (K+2) & (K+3)\\ 1 & 1 & 1\end{vmatrix}=0$$

$$\Rightarrow K=-2\qquad\Rightarrow|K|=2$$